



File: Power_Supply_1.kicad_sch



File: ADC_Channels.kicad_sch



File: Power_Supply_2.kicad_sch



File: Connectrors.kicad_sch



File: DevBuddy_Logic.kicad_sch

DevBuddy Mk1

PITHA INDUSTRIES

Sheet: /
File: DevBuddy.kicad_sch

Title: DevBuddy

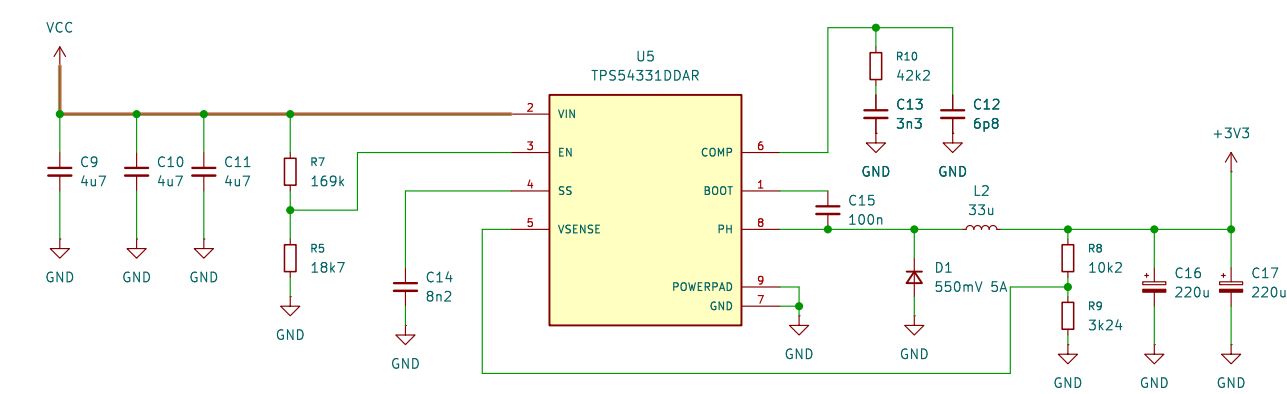
Size: A3 Date: 2026-02-08

KiCad E.D.A. 9.0.7

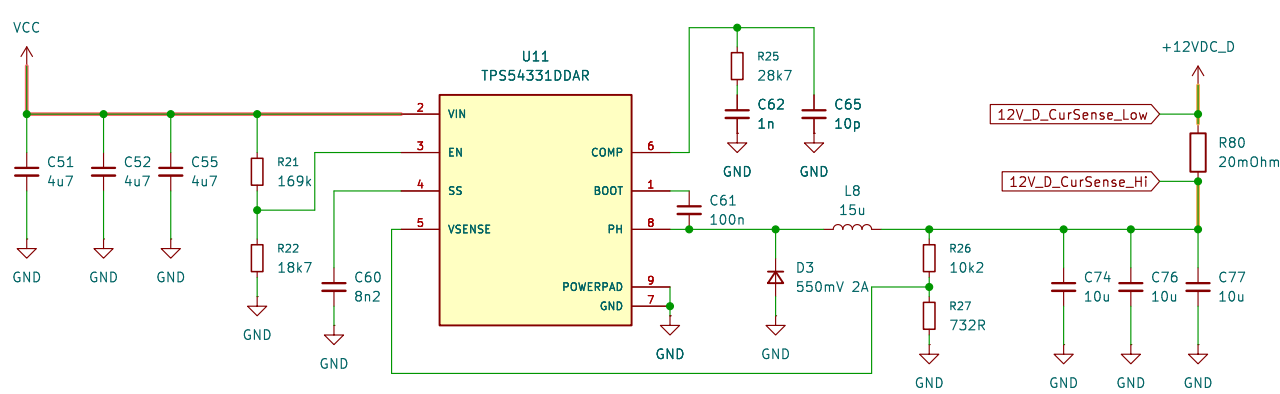
Rev: 1

Id: 1/6

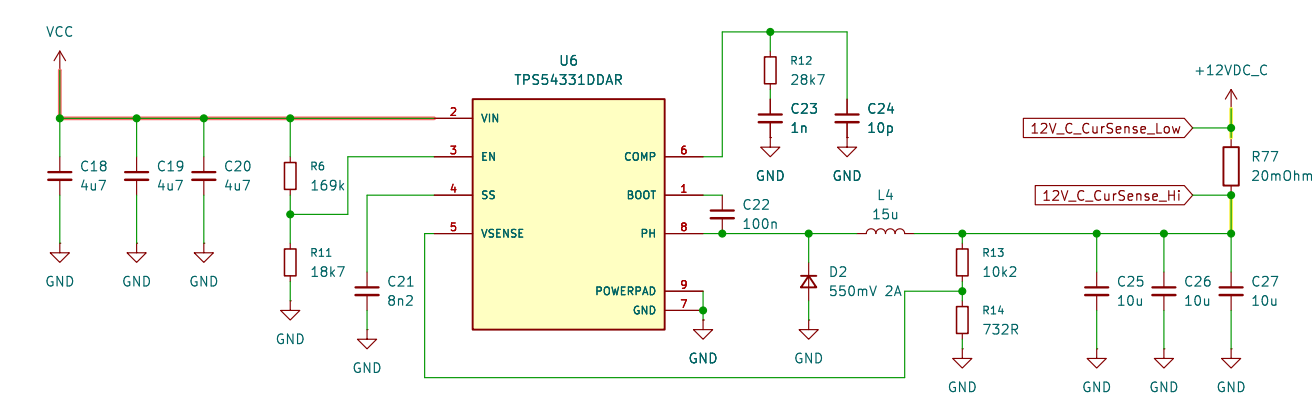
3V3 Logic Power supply @1A / 3.3W (could go up to 6.6W) CHANNEL A



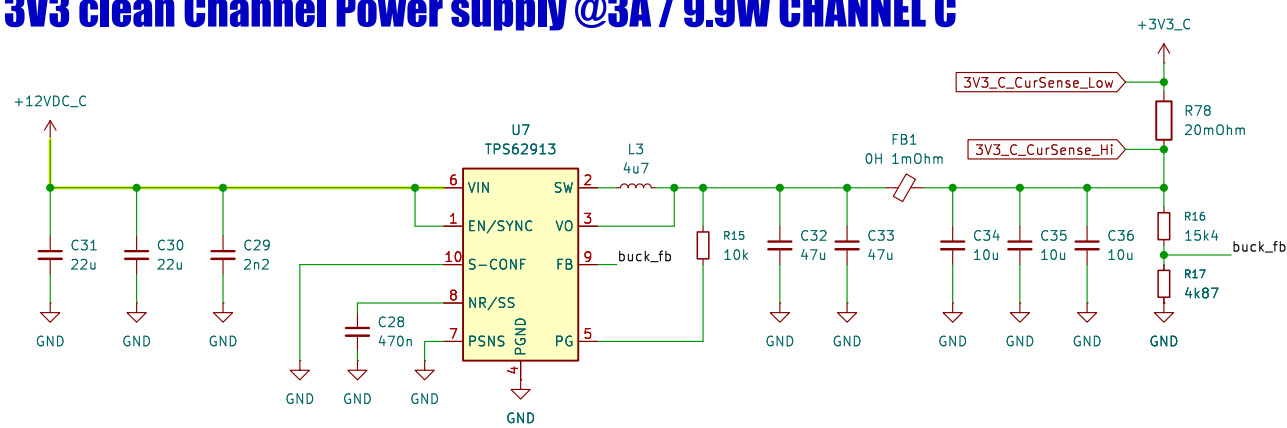
12V Supply for Low Voltage Clean Channels @2A / 24W CHANNEL D



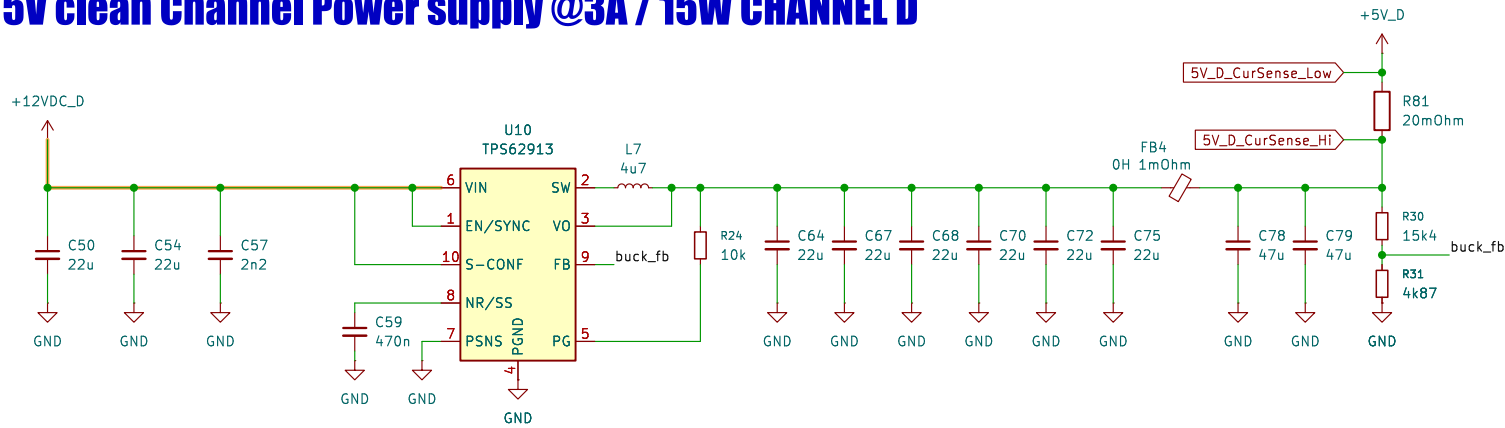
12V Supply for Low Voltage Clean Channels @2A / 24W CHANNEL C



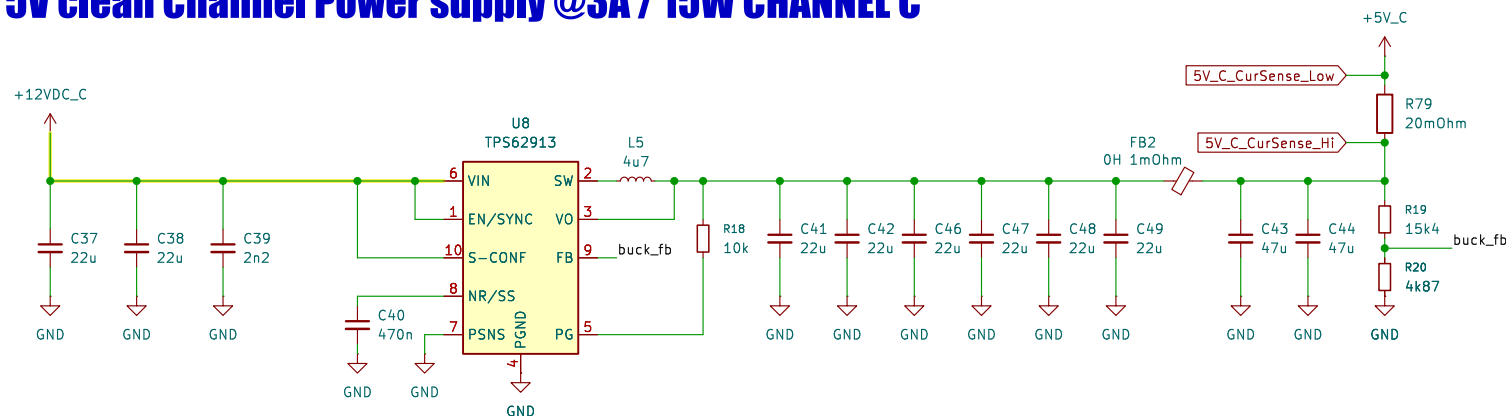
3V3 clean Channel Power supply @3A / 9.9W CHANNEL C



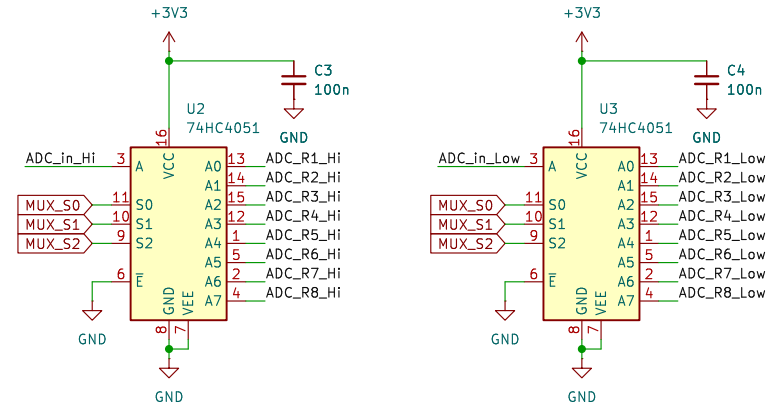
5V clean Channel Power supply @3A / 15W CHANNEL D



5V clean Channel Power supply @3A / 15W CHANNEL C

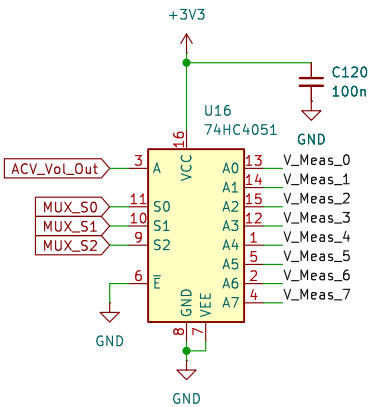


ADC Current MUX AND INA

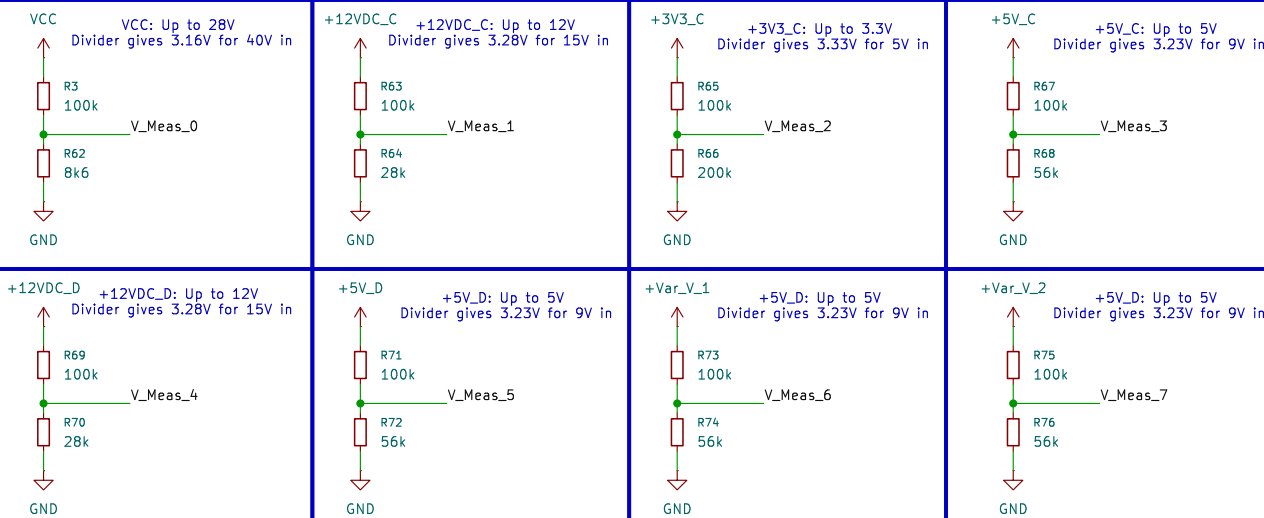


S2 S1 S0
L L L -> Y0 to Z
L L H -> Y1 to Z
L H L -> Y2 to Z
L H H -> Y3 to Z
H L L -> Y4 to Z
H L H -> Y5 to Z
H H L -> Y6 to Z
H H H -> Y7 to Z

ADC Voltage Mux



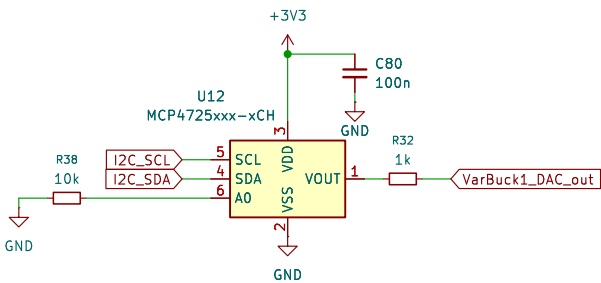
Safe voltage channels for MUX



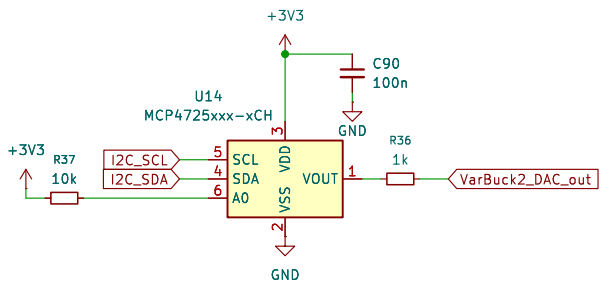
Current differential voltages

Vin	VCC_CurSense_Low	ADC_R1_Low
	VCC_CurSense_Hi	ADC_R1_Hi
12VDC_C	12V_C_CurSense_Low	ADC_R2_Low
	12V_C_CurSense_Hi	ADC_R2_Hi
3V3_C	3V3_C_CurSense_Low	ADC_R3_Low
	3V3_C_CurSense_Hi	ADC_R3_Hi
5V_C	5V_C_CurSense_Low	ADC_R4_Low
	5V_C_CurSense_Hi	ADC_R4_Hi
12V_D	12V_D_CurSense_Low	ADC_R5_Low
	12V_D_CurSense_Hi	ADC_R5_Hi
5V_D	5V_D_CurSense_Low	ADC_R6_Low
	5V_D_CurSense_Hi	ADC_R6_Hi
Var_V2	Var_V2_CurSense_Low	ADC_R7_Low
	Var_V2_CurSense_Hi	ADC_R7_Hi
Var_V1	Var_V1_CurSense_Low	ADC_R8_Low
	Var_V1_CurSense_Hi	ADC_R8_Hi

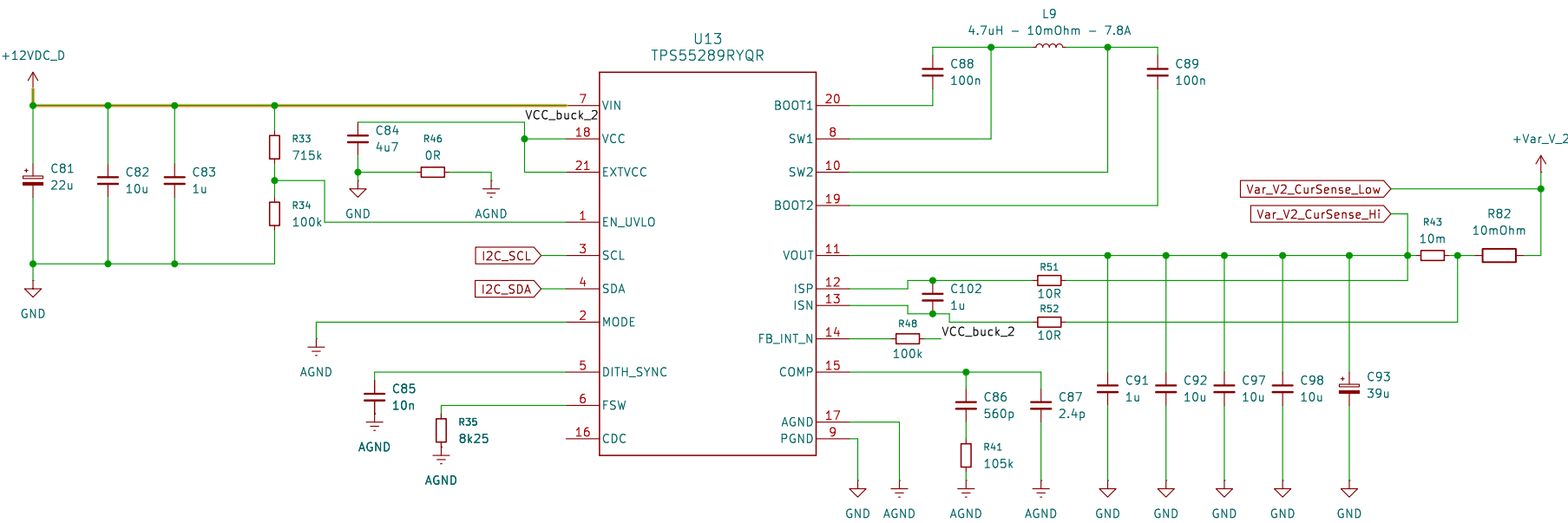
DAC1 controlling output voltage 1



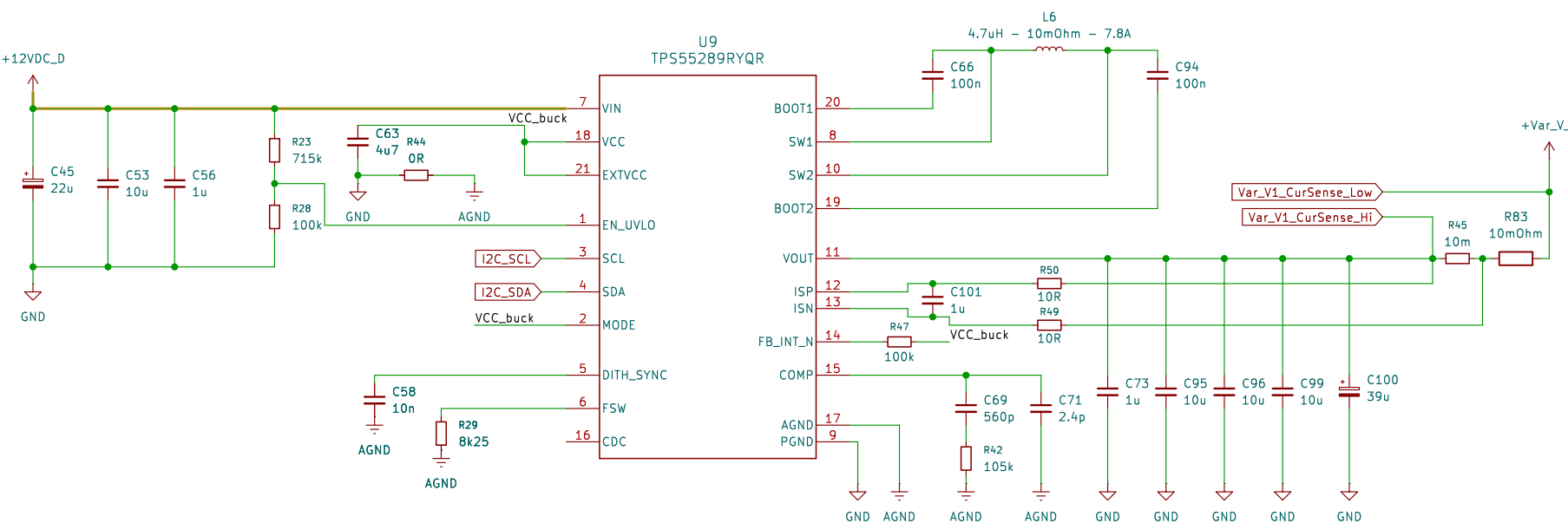
DAC2 controlling output voltage 2



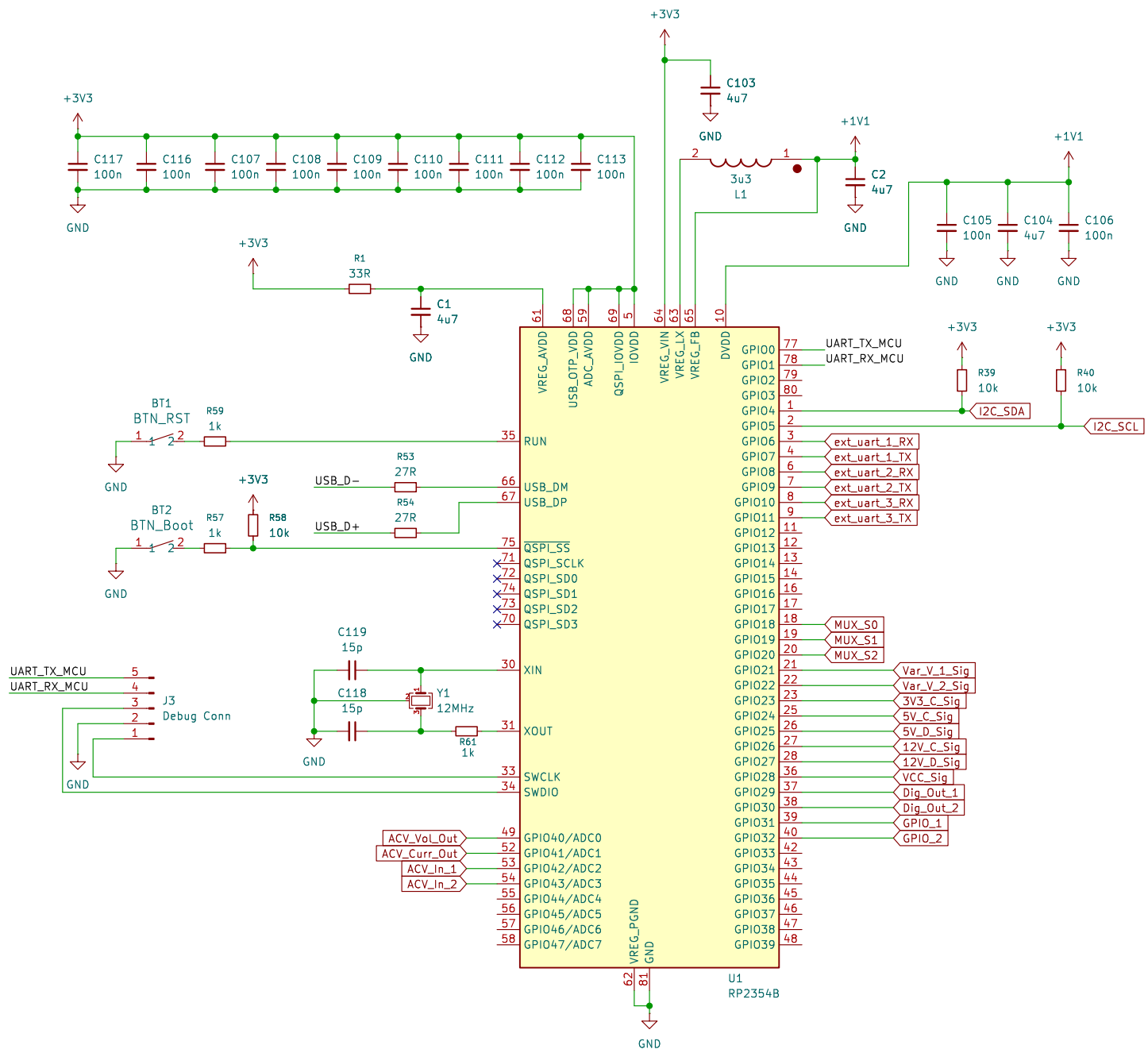
VarBuck2: Variable Clean Voltage from 0.8V up to 3.8V+ | Addr 74h



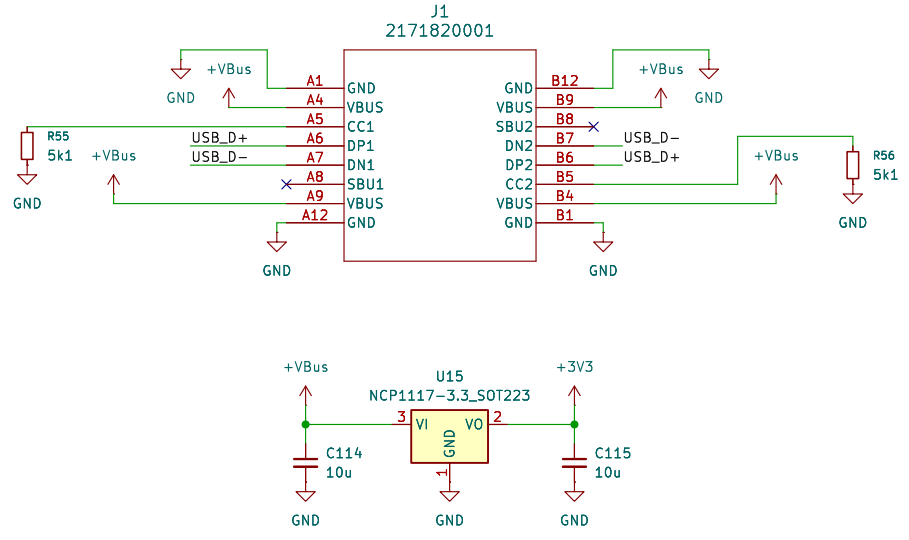
VarBuck1: Variable Clean Voltage from 0.8V up to 3.8V+ | Addr 75h



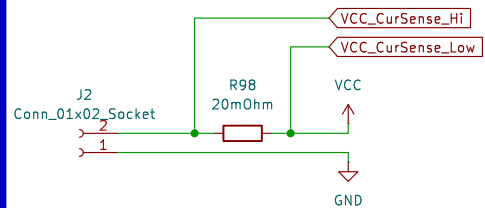
RP2354B Design



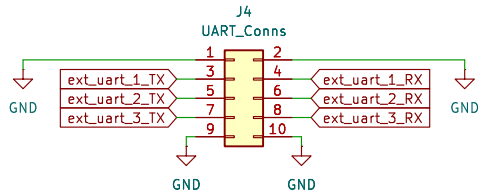
USB C and USB logic power suuply



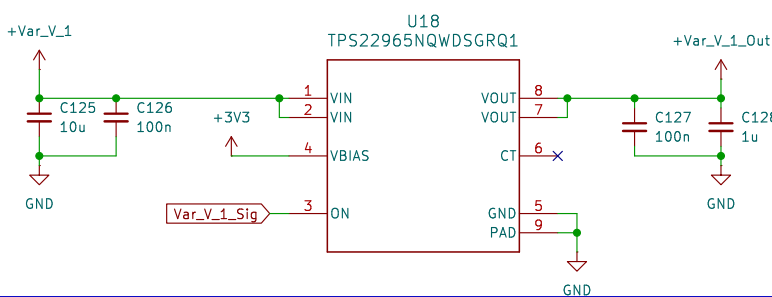
Input power



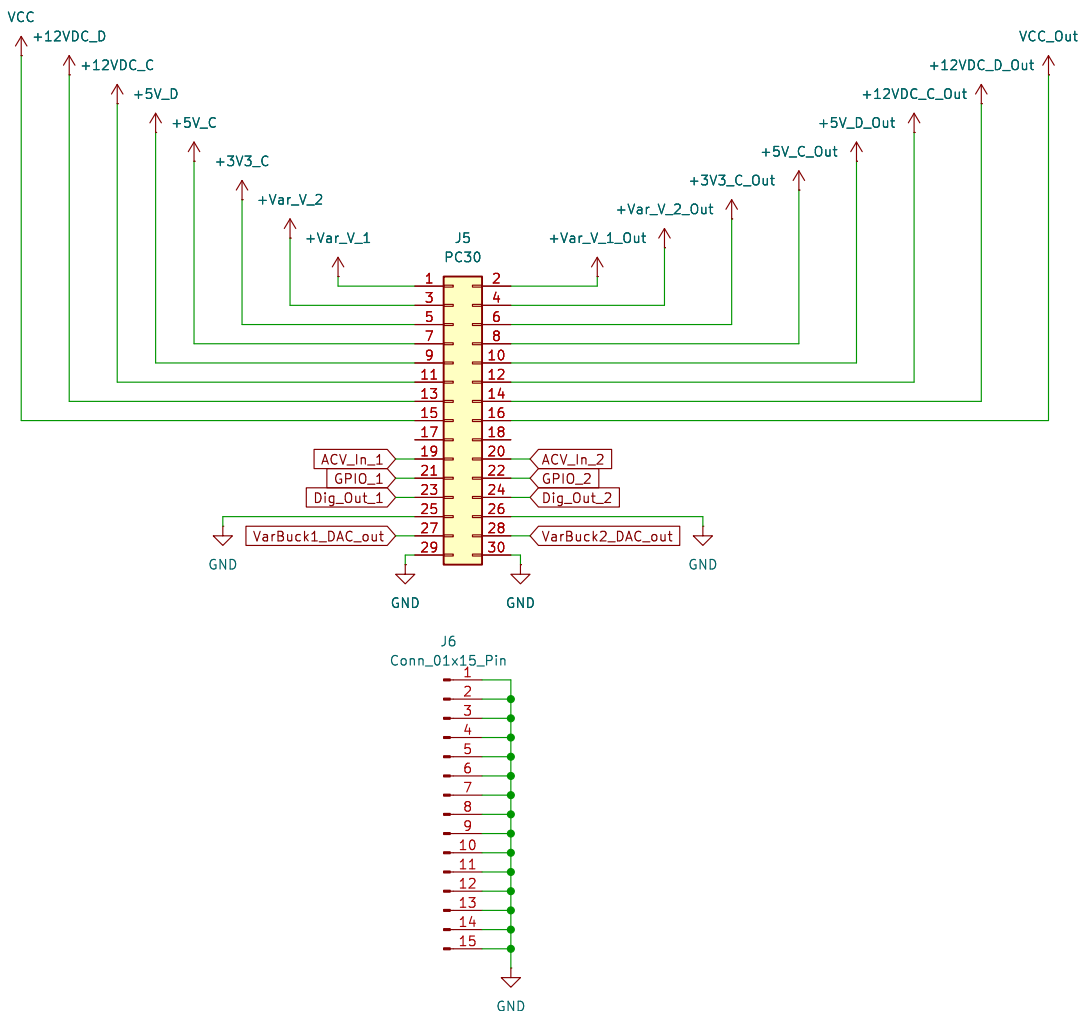
Uart conns



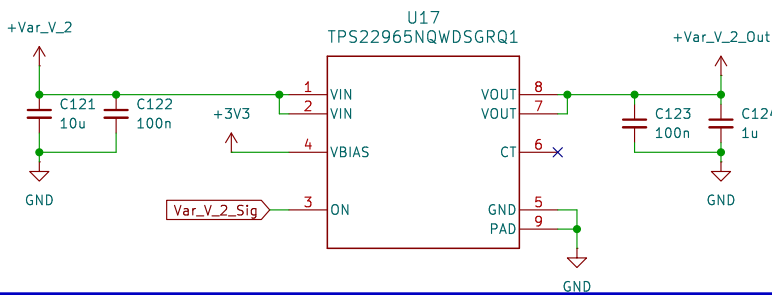
Var_V_1_Out



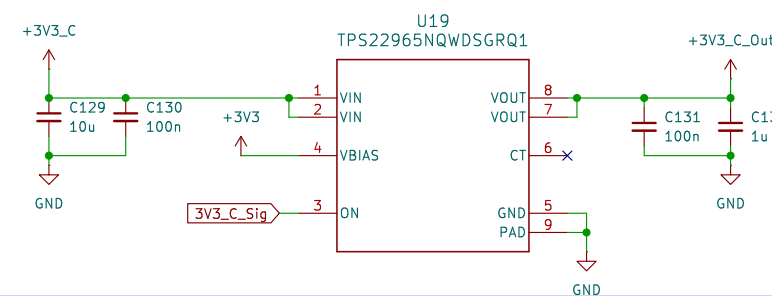
PC30-PC15 Main Connectors



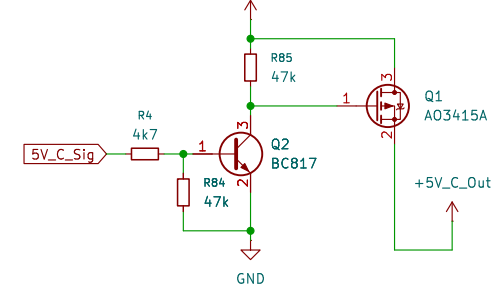
Var_V_2_Out



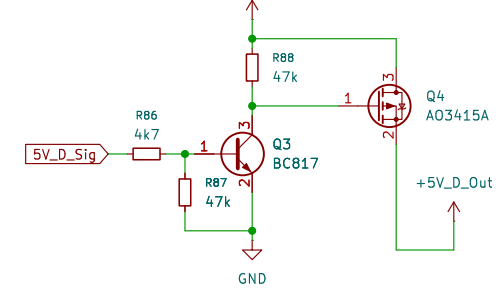
3V3_C_Out



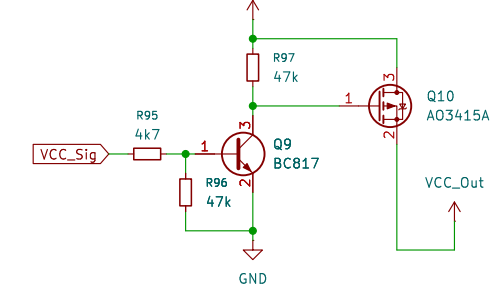
5V_C_Out



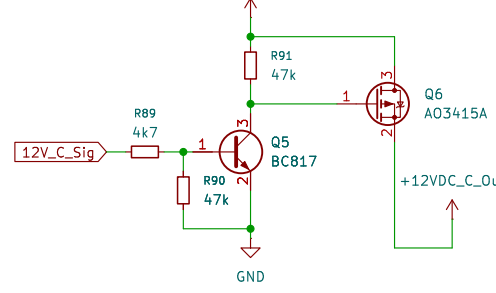
5V_D_Out



VCC_Out



12V_C_Out



12V_D_Out

